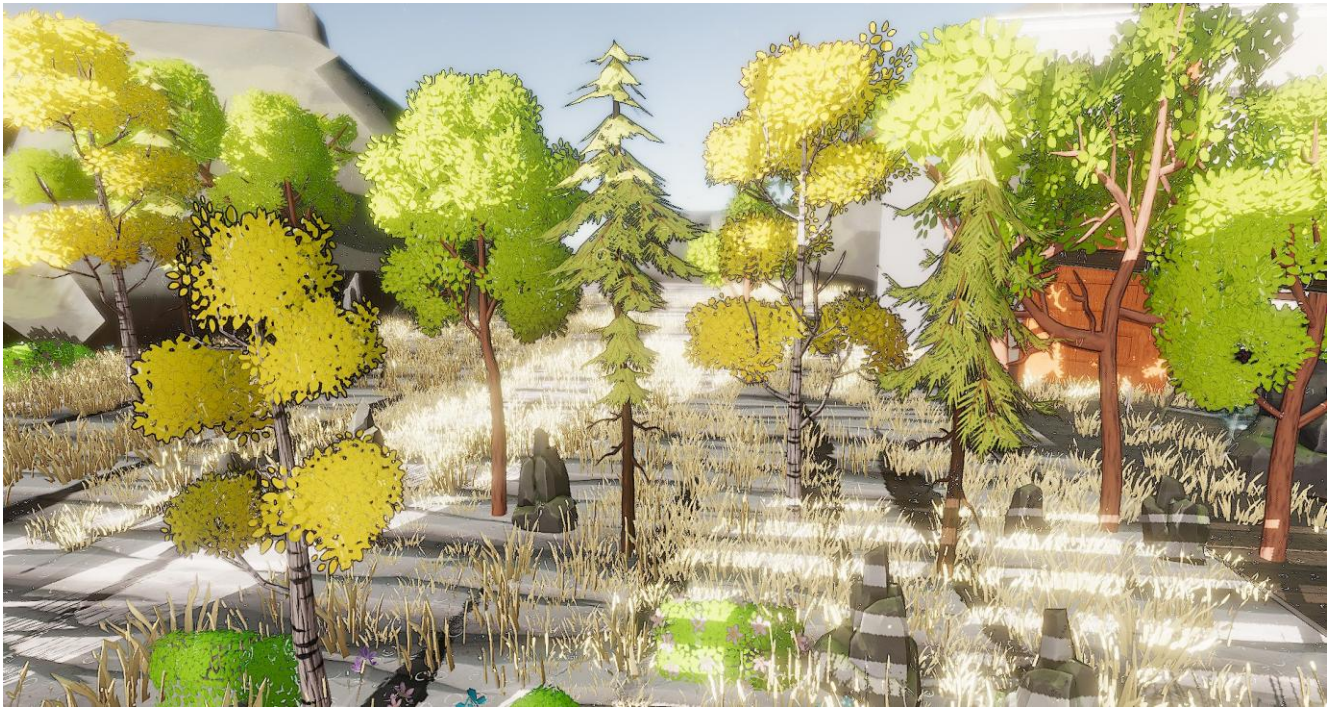


GLAMOR Image Effect Stack Framework



GLAMOR v2.0



GLAMOR URP Setup Guide

Overview

The **GLAMOR Image Effect Stack Framework** system is a next generation image effects framework for Unity URP.

The system is compatible with [Sky Master ULTIMATE URP](#) version, and can be combined with the **Ethereal** module to add volumetric lighting to further enhance the look.

The system also includes a screen space sun shafts effect to quickly add volume to the atmosphere when the sun is inside or near the camera field of view.

GLAMOR can be upgraded with a big discount to the [Environment Building Bundle](#) that includes **Sky Master ULTIMATE** and all other major [ARTnGAME](#) assets.

For quick start on the main system, please refer to the tutorial video playlist in the following link:
[GLAMOR Tutorial Playlist](#)

Video playlist showcasing the use of GLAMOR with Sky Master ULTIMATE URP Volumetric Lighting & Cloud and LUMINA Real Time global illumination.

[GLAMOR with Sky Master ULTIMATE URP Video](#)

For any questions please contact me in ARTnGAME discord channel:
<https://discord.gg/X6fX6J5>

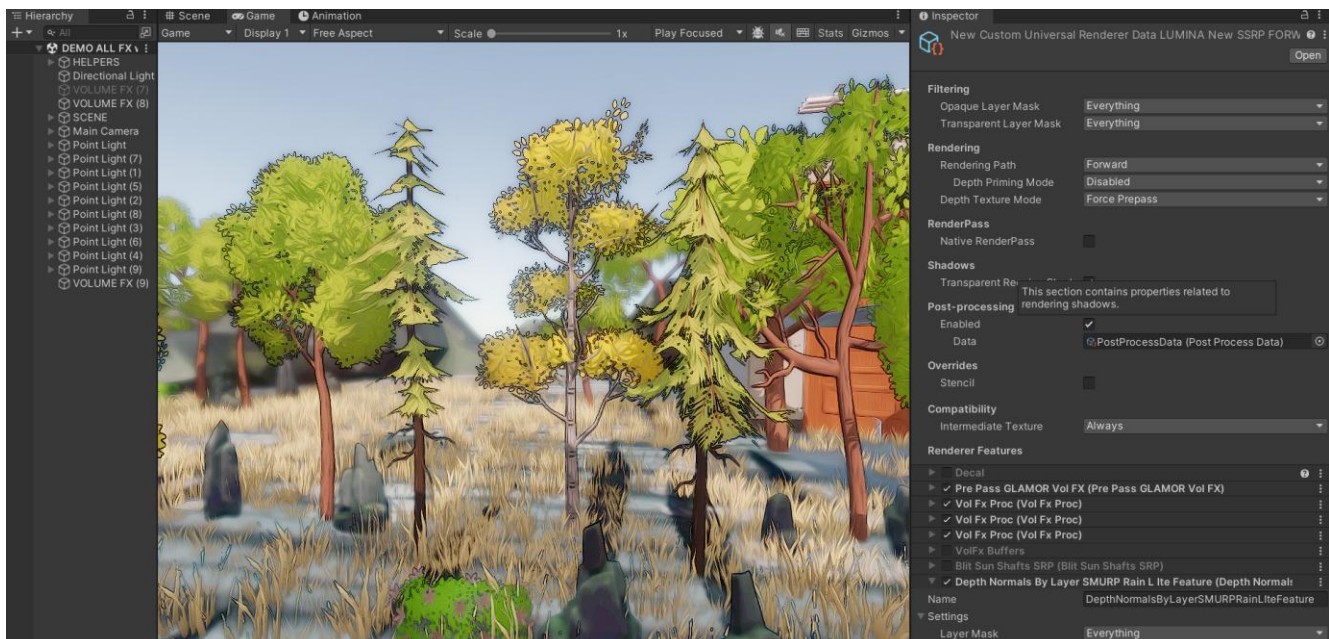
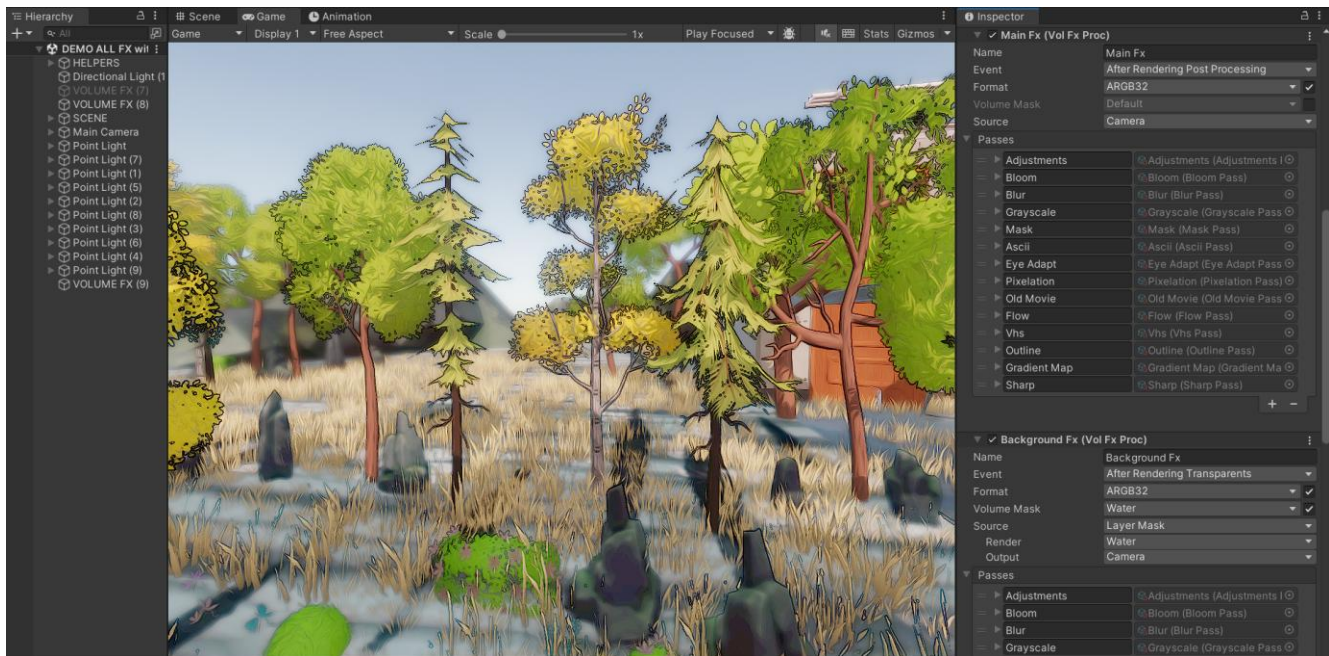
Or in ARTnGAME email:
artengames@gmail.com

GLAMOR URP Setup steps

The system setup requires the URP pipeline to have been setup and **two forward renders** assigned to the pipeline Renders list, **one for the GLAMOR effects** and **one for the Depth Normals texture creation**, used for the Rain FX.

It is advised to add a third renderer without any image effects, that will be used for the case where other helper cameras need to be rendered without effects applied. This third renderer can be set as first default option, so helper cameras that automatically grab the default renderer are not affected.

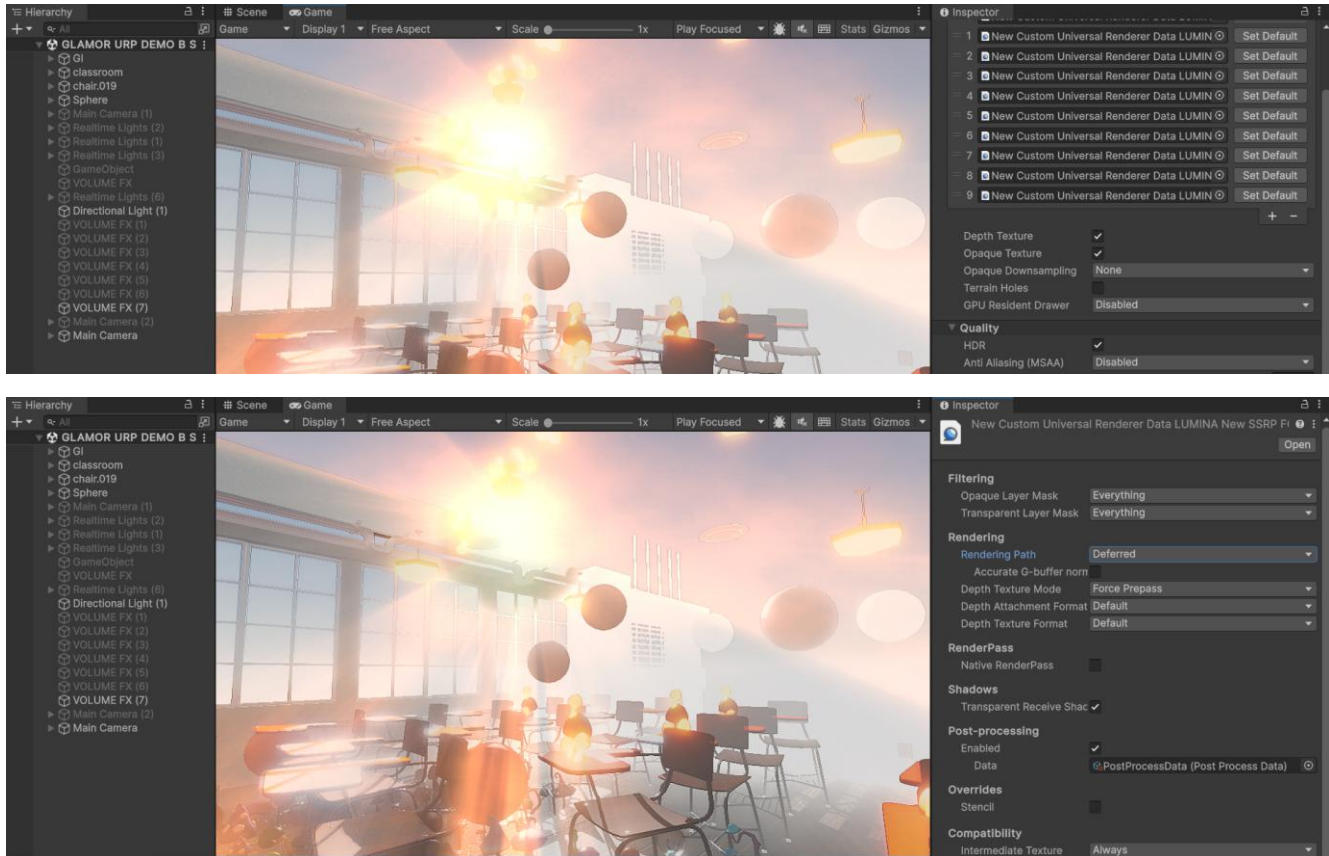
The effects are setup by adding the “VolFxFxProc” renderer feature in the renderers and **then add effects in its stack** as shown in the following images. Can **choose between affecting the full screen** by Camera option in Source **or select layers to affect** by the Layer Mask option in Source.



IMPORTANT: For use with Unity 6 or Unity 2023.3 Render Graph, a special initializer renderer feature named “PrePassGLAMORVolFX” must be inserted in the renderer and set to after image effects in render order, in order for the framework to function.

The “**Depth Texture**” and “**Opaque Texture**” checkboxes in the URP pipeline main settings must be enabled. Also **Linear color space** must be used, as the system is configured for that mode.

IMPORTANT: In some cases and Unity versions it is required to either enable or disable the 2x AA in URP pipeline settings, for the color buffer to be available correctly from Unity. In Unity 2023.3 can use any of Forward, Deferred and Forward+ and remove the 2x AA.



IMPORTANT: Make sure the scene camera is tagged as “MainCamera” for the effects to function properly.

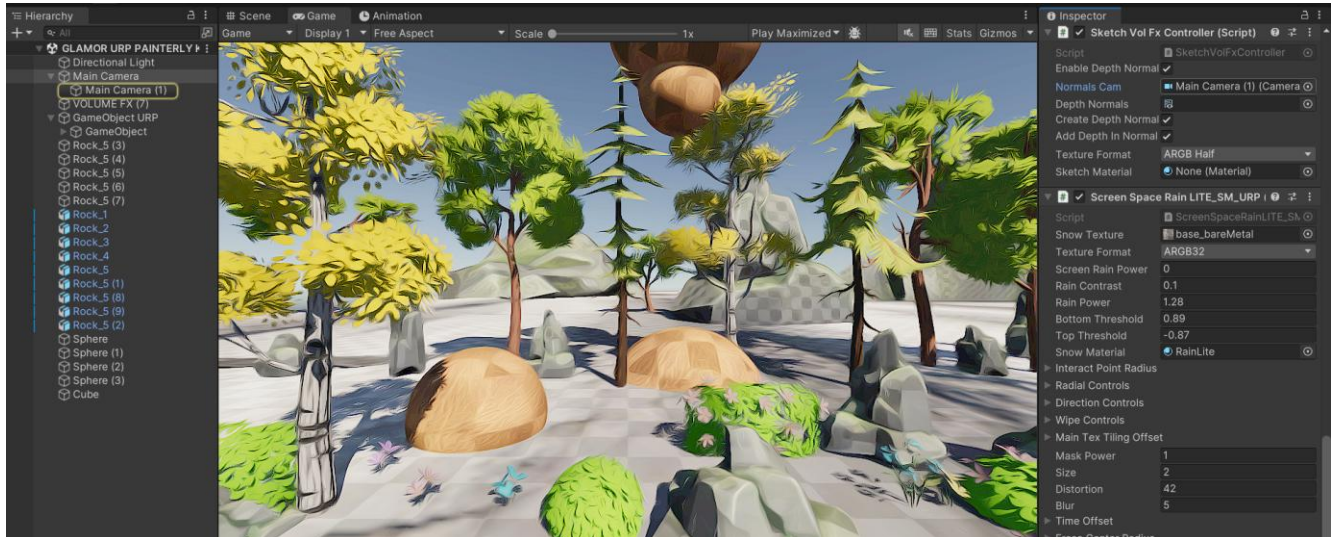
IMPORTANT: (Materials used in the Effects)

The various effects that are controlled only in the Volume create a material instance per volume directly from the effect shader. The more complex effects that require an extra separate controller than the Volume, use in order, **first the material defined in the Volume entry of the effect**, if this is not defined looks for a pre-defined material name in the Resources that implements the effect shader. These effects are the Rain, Fog and Sun Shafts ones.

IMPORTANT: (Depth Normals Texture)

In the non Render Graph mode, using a second camera for the Depth Normals texture can be avoided by adding the Depth Normals renderer feature in the renderer of the Main Camera. The renderer feature is named as “**DepthNormalsByLayerSMURPRainLiteFeature**”.

In the case that two controllers that support control of the Depth Normals camera are used, only one need to be setup with the camera and will create the texture as needed for use for the other effects.

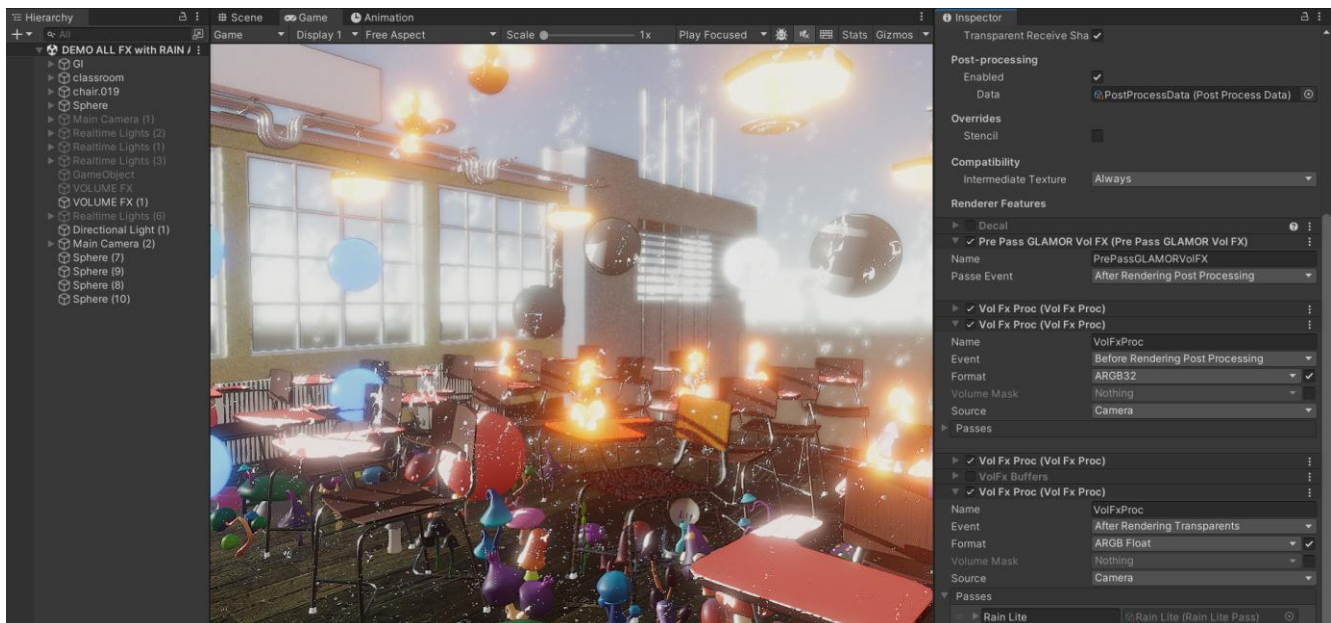


GLAMOR Effects – Advanced FX List

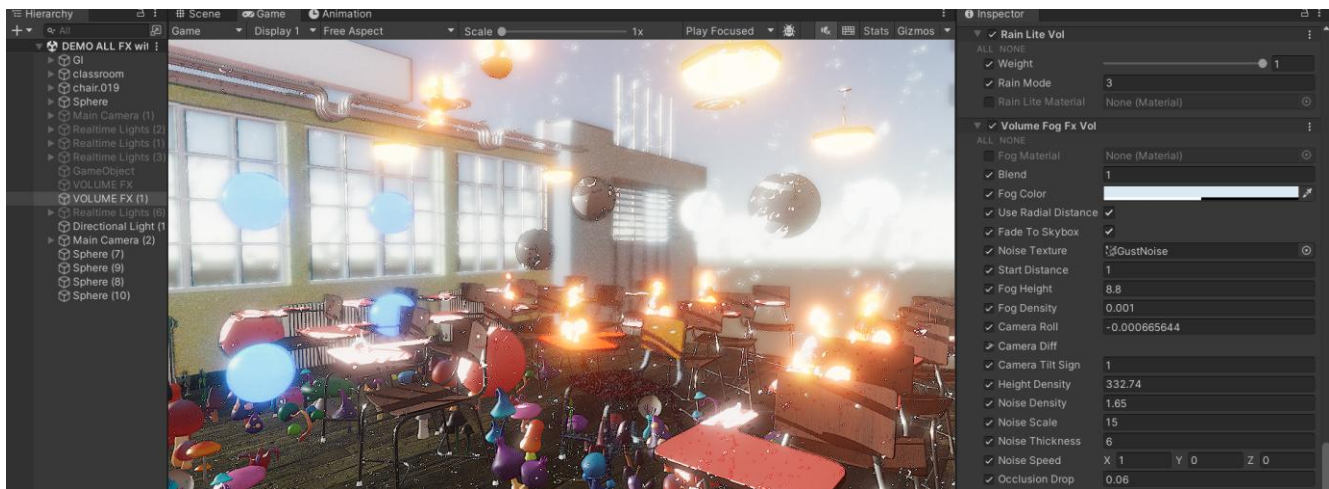
The advanced effects included in the system require extra controller(s) outside the Volume ones and extra helper textures e.g. DepthNormals texture.

1. RAIN EFFECT

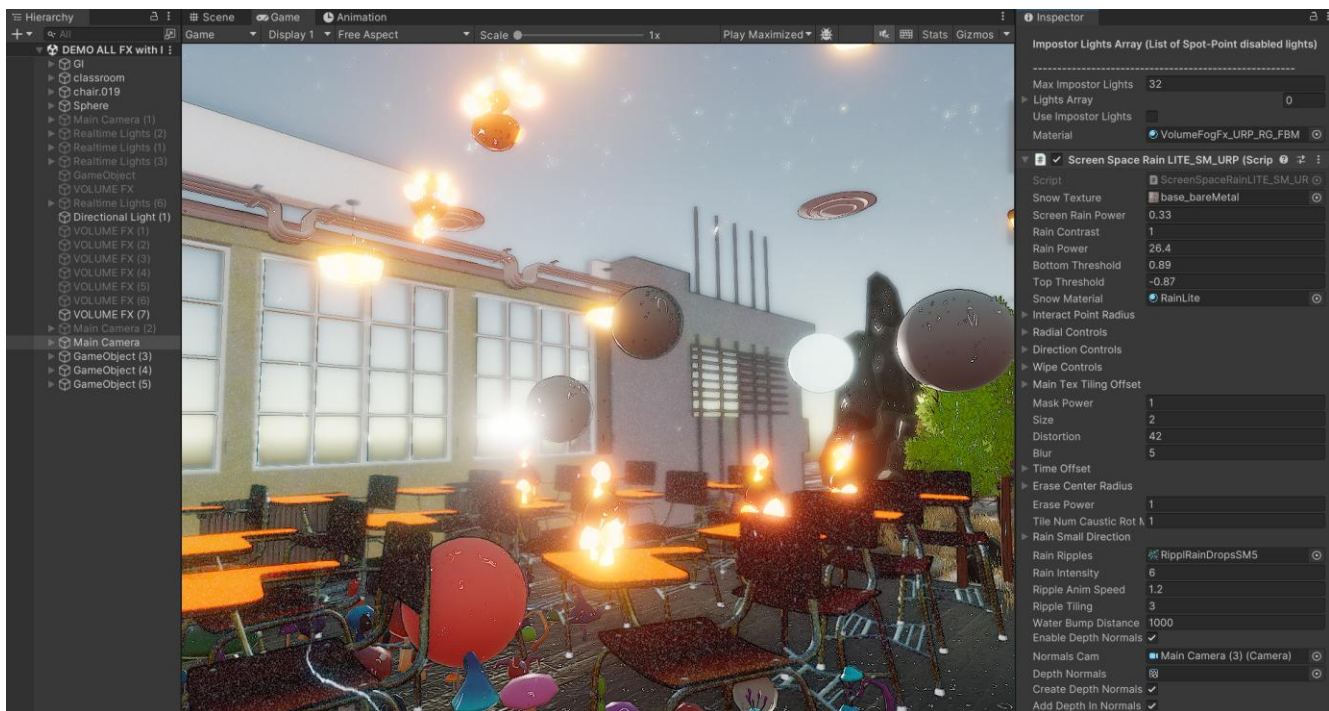
The setup of the rain in the pipeline is done by adding the “**Rain Lite**” effect entry in the “**VolFxProc**” stack renderer feature.



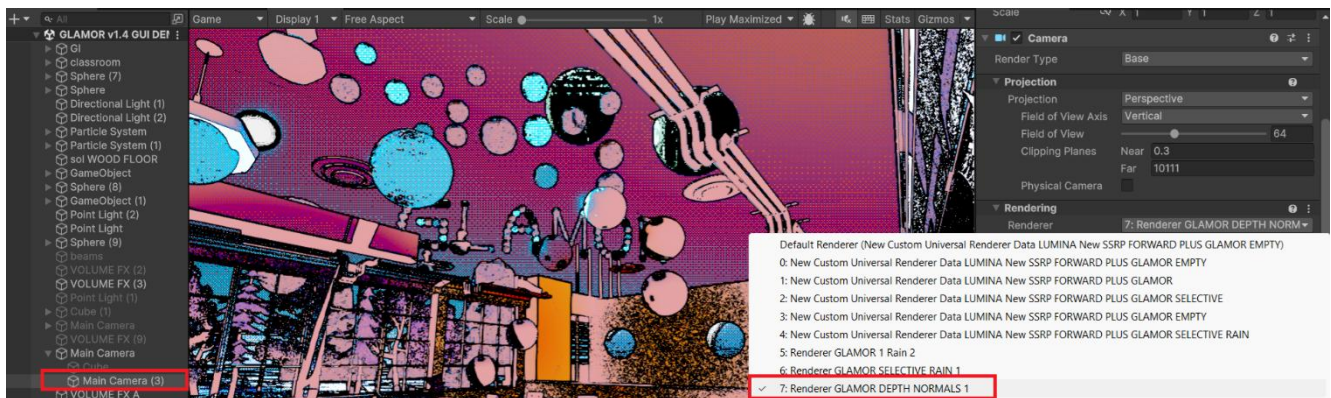
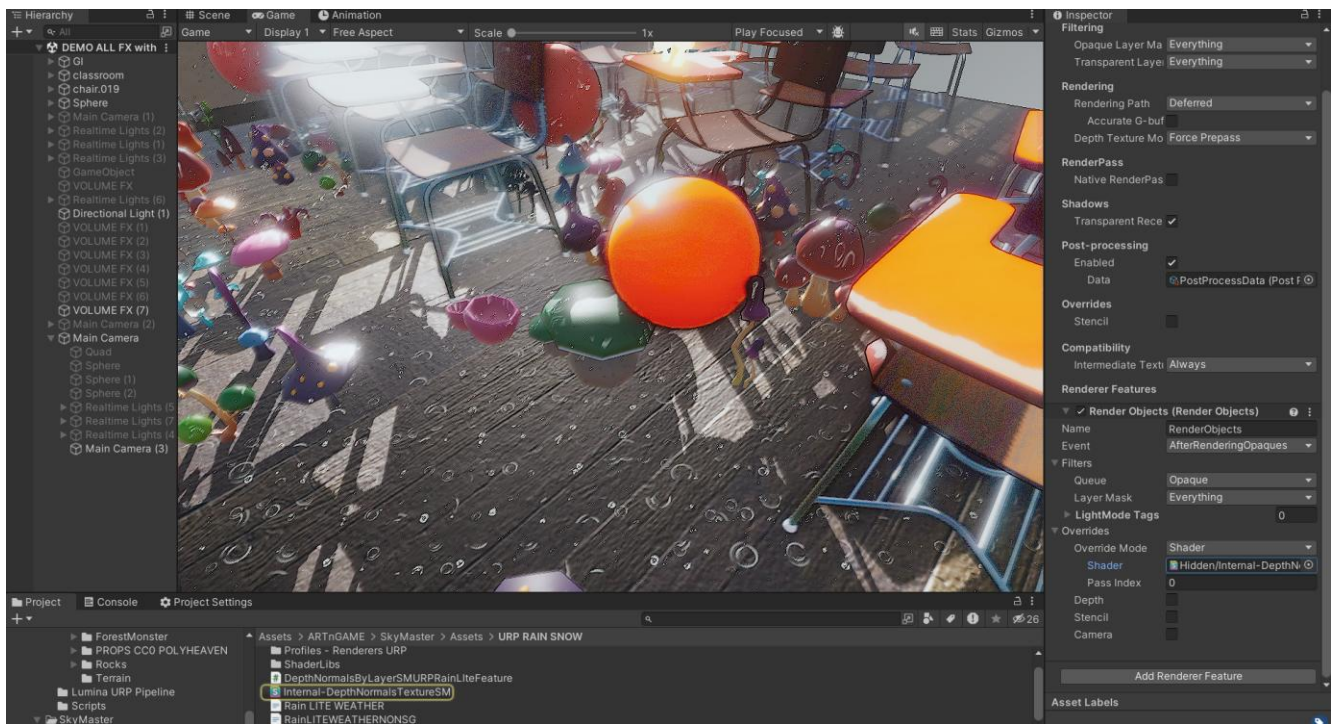
The setup of the rain in the scene is done by adding an entry in one of the scene volumes, local or global, as shown below.



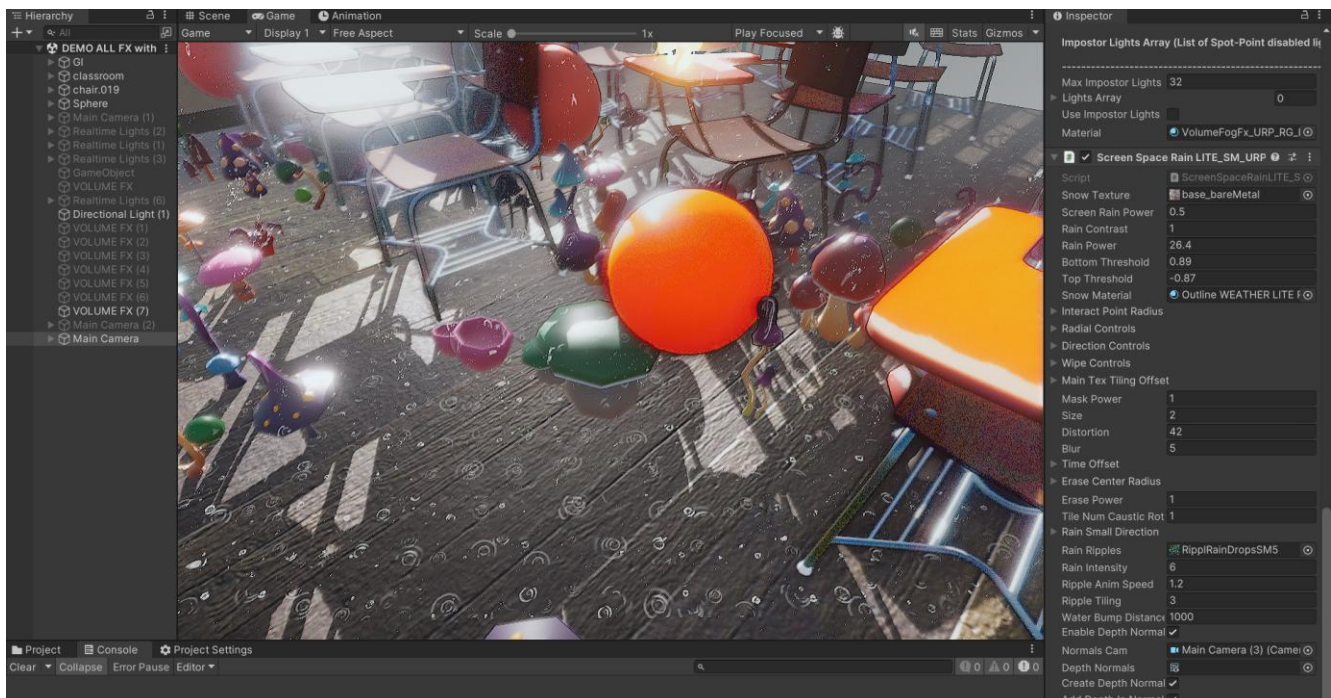
The control of the effect is done by placing the “**ScreenSpaceRainLITE_SM_URP_FX**” script in the Main Camera in the scene. The controller controls directly the referenced material in the “**Snow Material**” slot, this material can be set in the Rain entry in the scene Volume “**Rain Lite Vol**” in “**Rain Lite Material**” slot, so can correspond to the controlled material.



For the rain to be ground on the scene objects, a second camera is used in the scene, parented with zero translation and rotation to follow the main camera and uses one renderer configured with the below setup of “**Render Objects**” renderer feature, using the “**Internal-DepthNormalsTextureSM**” Shader to produce the Depth Normals texture of the scene.

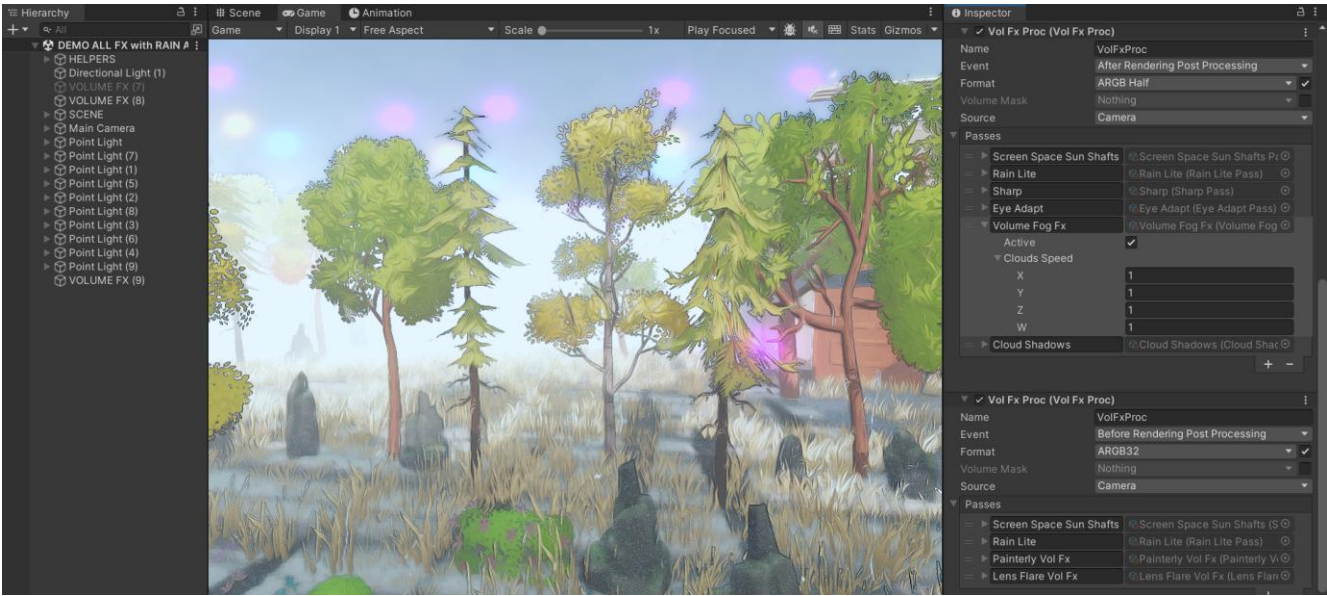


The **Depth Normals** camera must be referenced in the Rain controller, as shown below, so can be auto configured for the render. [Video](#) of the Depth Normals configuration procedure for reference.

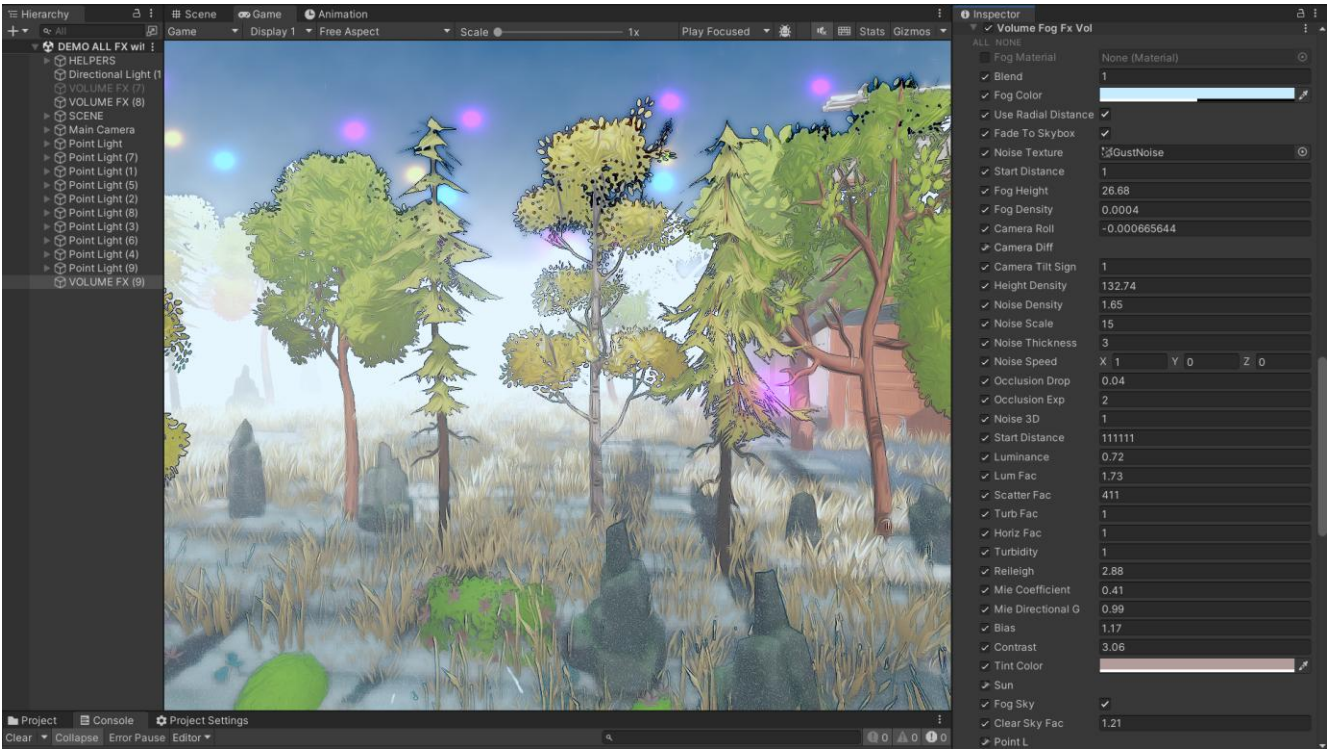


2. VOLUMETRIC FOG

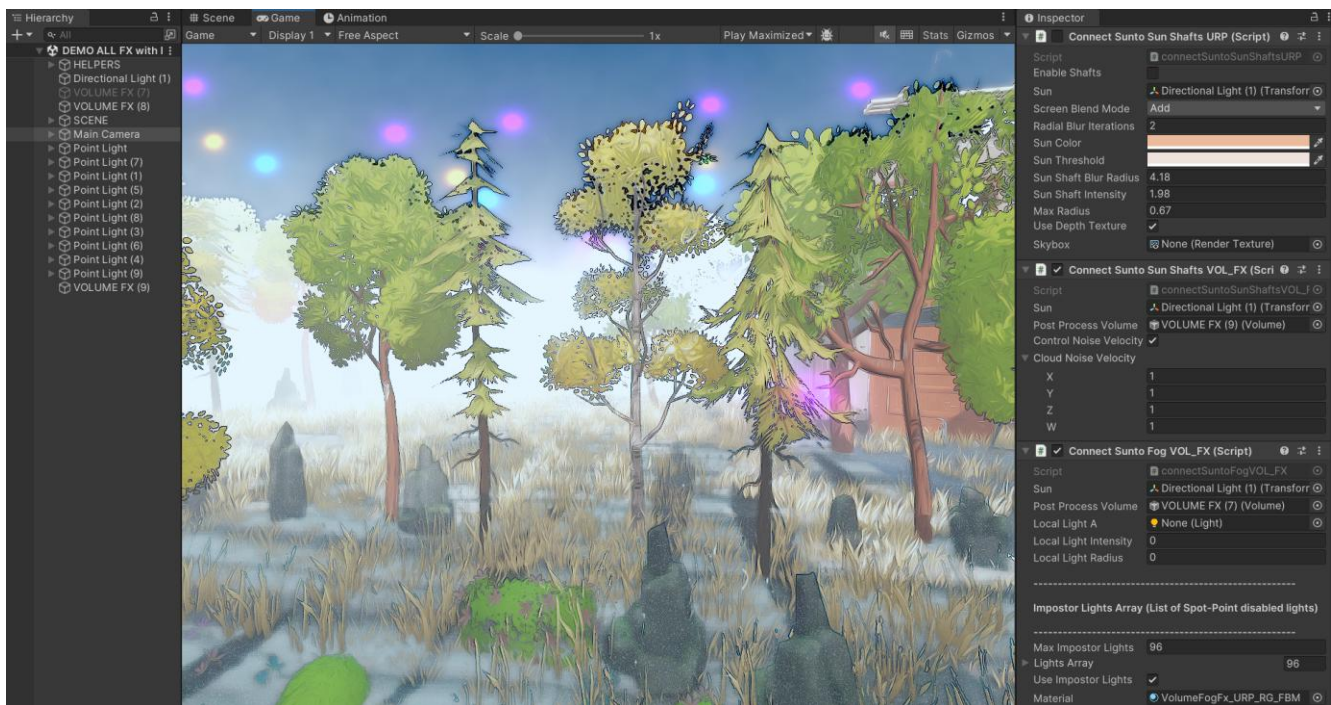
The setup of the Volumetric Fog in the pipeline is done by adding the “Volume Fog Fx” effect entry in the “VolFxProc” stack renderer feature.



The setup of the fog in the scene is done by adding an entry in one of the scene volumes, local or global, as shown below.



The setup of the **Volumetric Fog** system controller is done by placing the “**connectSuntoFogVOL_FX**” script in the Main Camera in the scene.

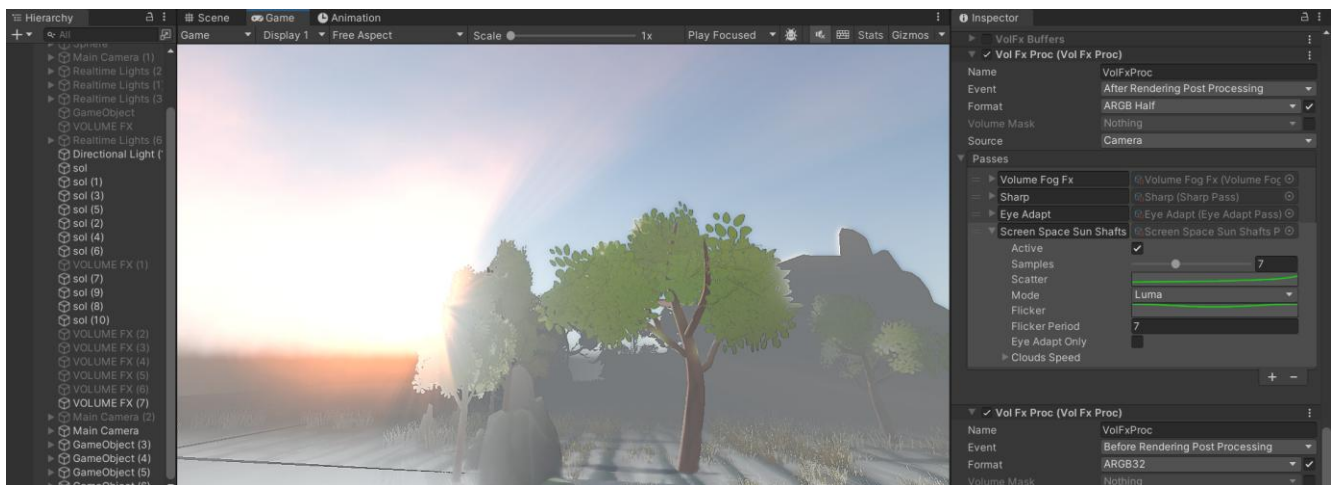


The system supports impostor light volumes, by adding point or spot lights, enabled or disabled in the scene, in the “**Lights Array**” list. Make sure to set the number of lights also in the Max Impostor Lights value, in order for the volumes to be enabled. Enable the “**Use Impostor Lights**” checkbox to activate the effect.

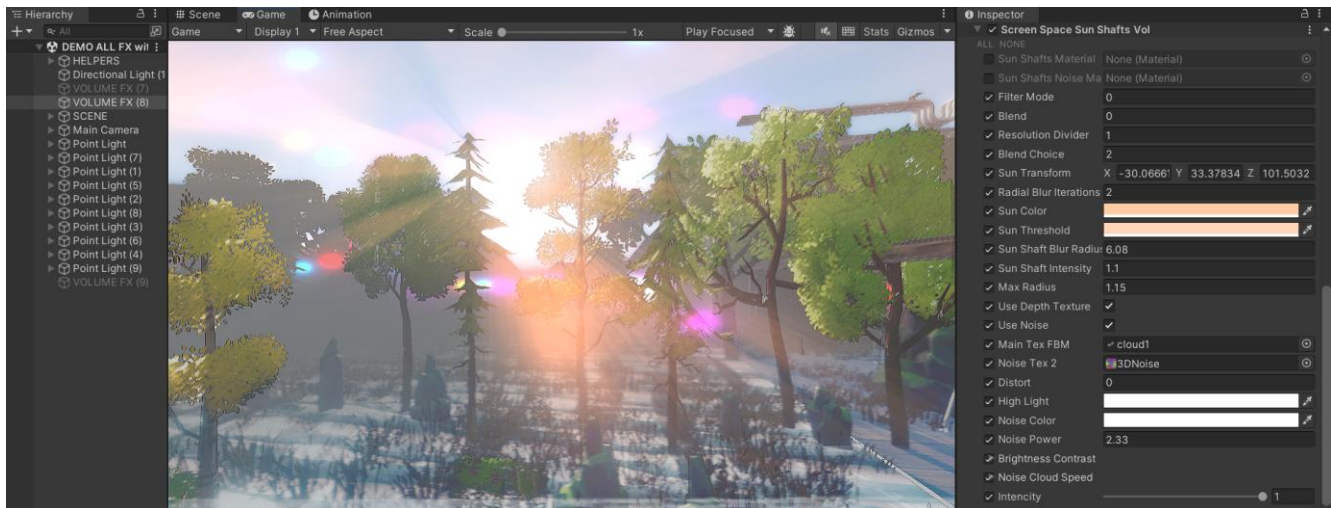
[Video](#) of tweaking the Volume Fog for height fog control.

3. SUN SHAFTS

The setup of the **Sun Shafts** in the pipeline is done by adding the “**Volume Fog Fx**” effect entry in the “**VolFxProc**” stack renderer feature.

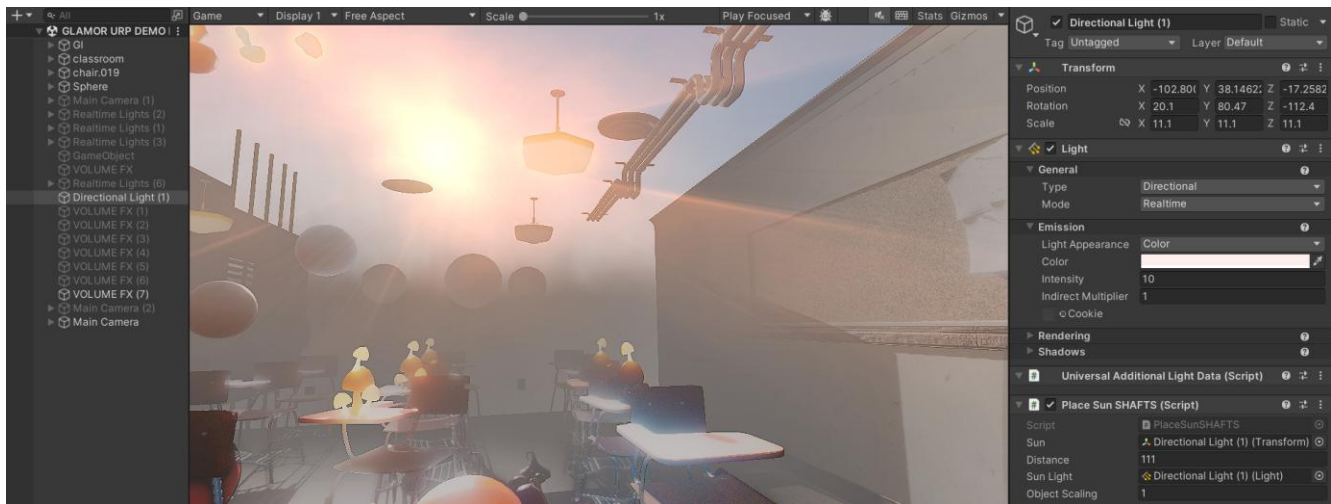


The setup of the sun shafts in the scene is done by adding an entry in one of the scene volumes, local or global, as shown below.

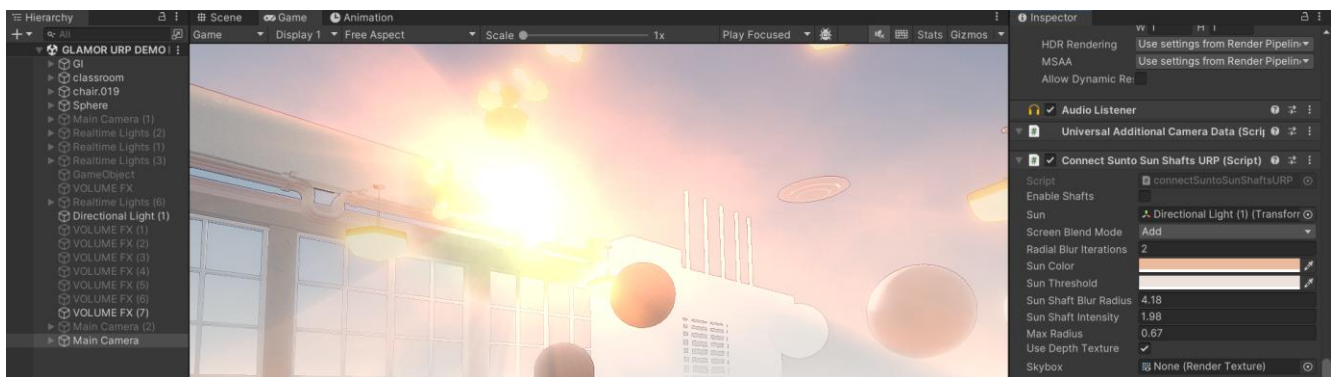
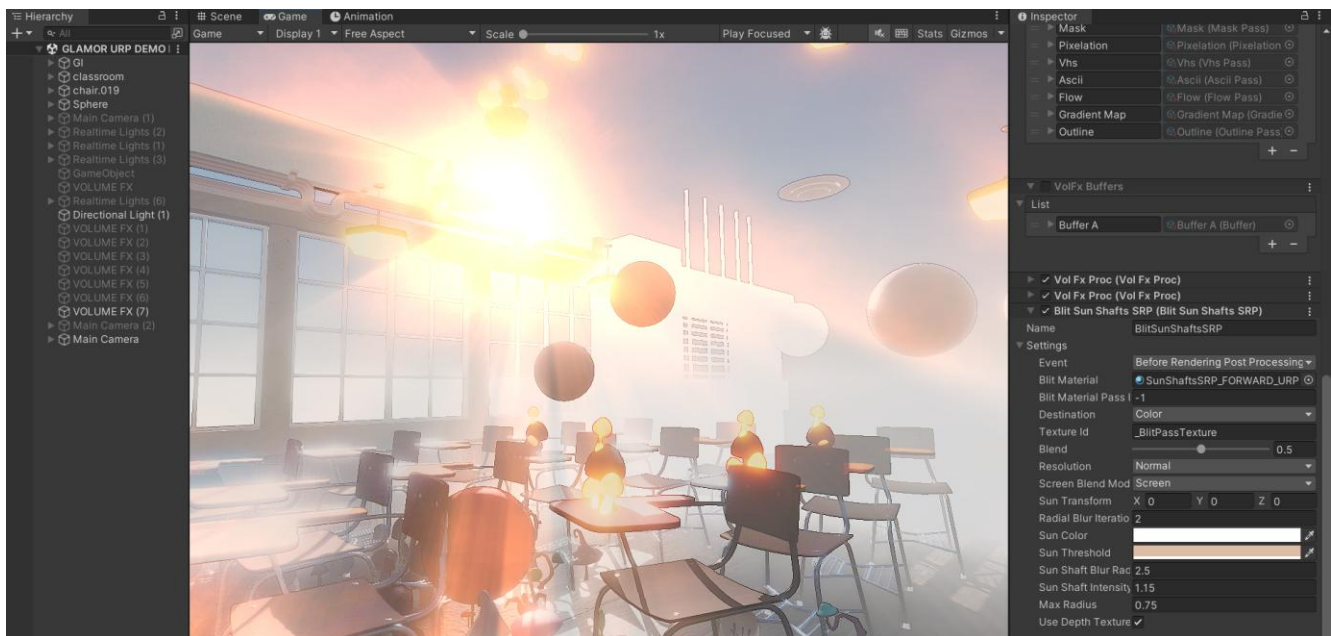


The setup of the sun shafts system controller is done by placing “connectSuntoSunShaftsVOL_FX” script in the Main Camera in the scene.

The “PlaceSunSHAFTS” script must be added in the sun light, in order to place its position correctly in the sky based on the light rotation. This is important as the sun shafts require the true position of sun in the sky to function.



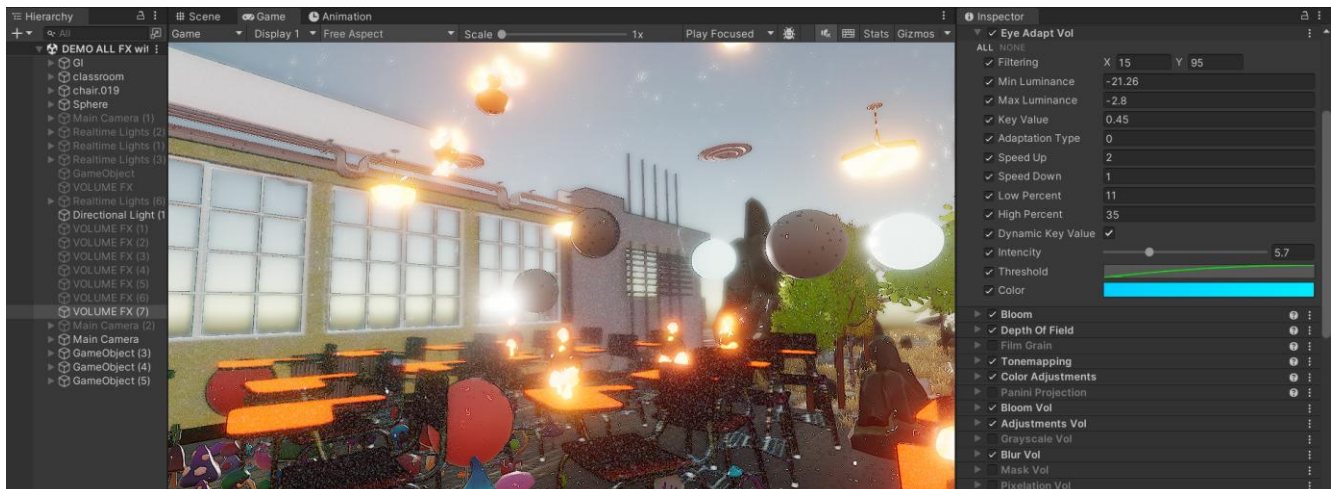
There is also an experimental standalone version of the effect that uses the “connectSuntoSunShaftsURP” controller in the camera and “BlitSunShaftsSRP” renderer feature in the camera renderer. It is advised to use the volume integrated system instead, as is more developed and integrates better with the other effects in the stack.



4. AUTO EYE EXPOSURE

The setup of the **Auto Eye Exposure Adaptation** filter in the pipeline is done by adding the **"EyeAdaptVolFx"** effect entry in the **"VolFxProc"** stack renderer feature.

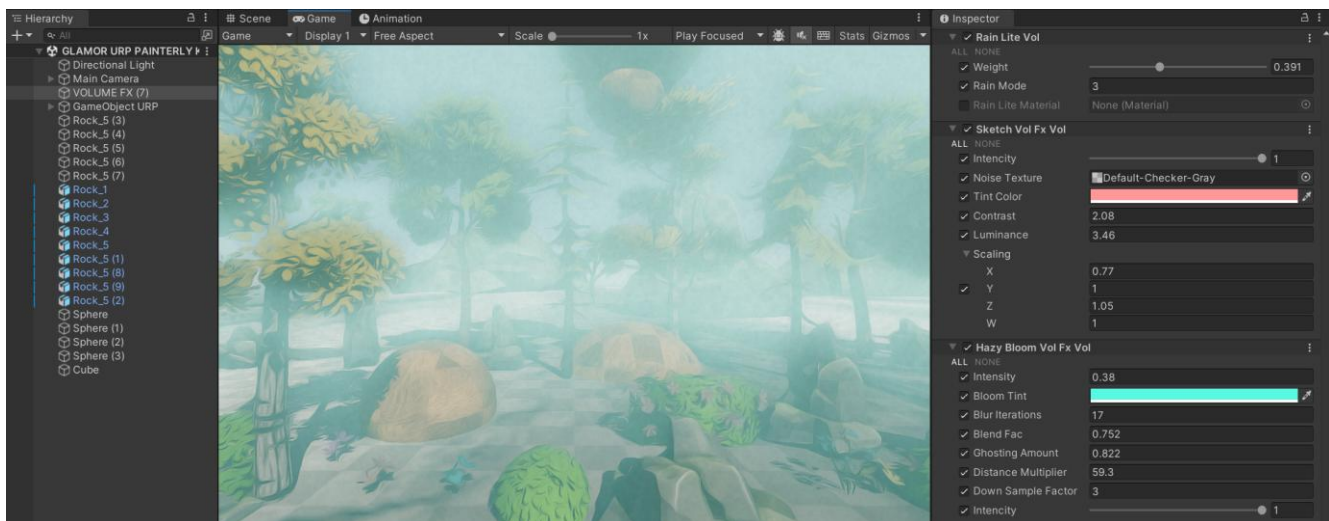
The setup of the effect in the scene is done by adding an **"EyeAdaptVol"** entry in one of the scene volumes, local or global, as shown below.



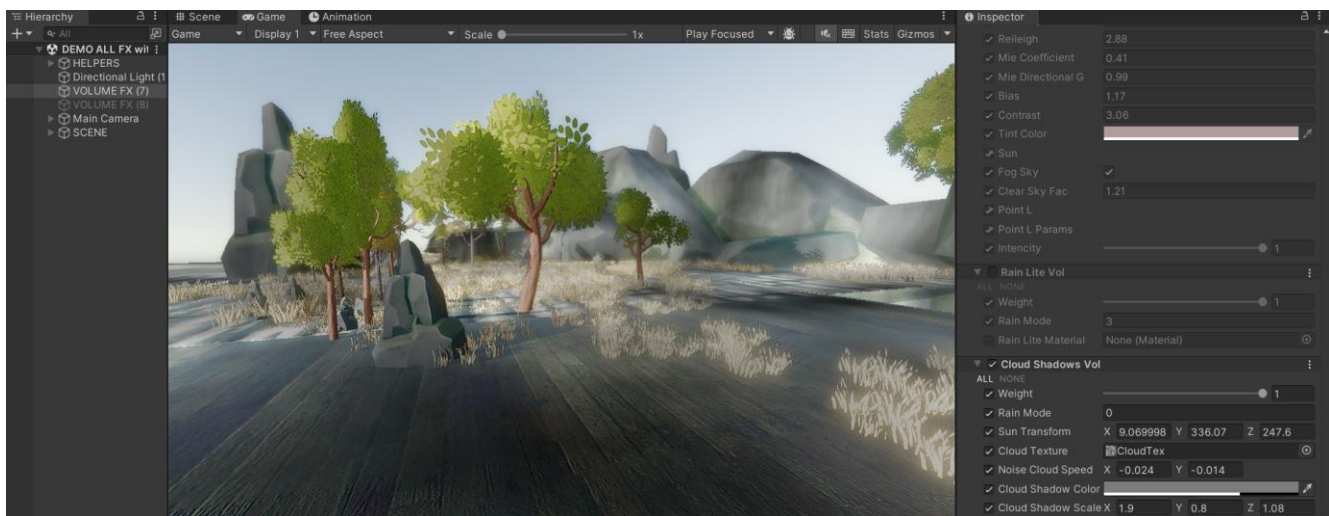
GLAMOR Effects List in Volume

The system has an extensive range of effects that are directly controlled in the Volume, after enable them in the pipeline renderer, using the “VolFxProc” entries.

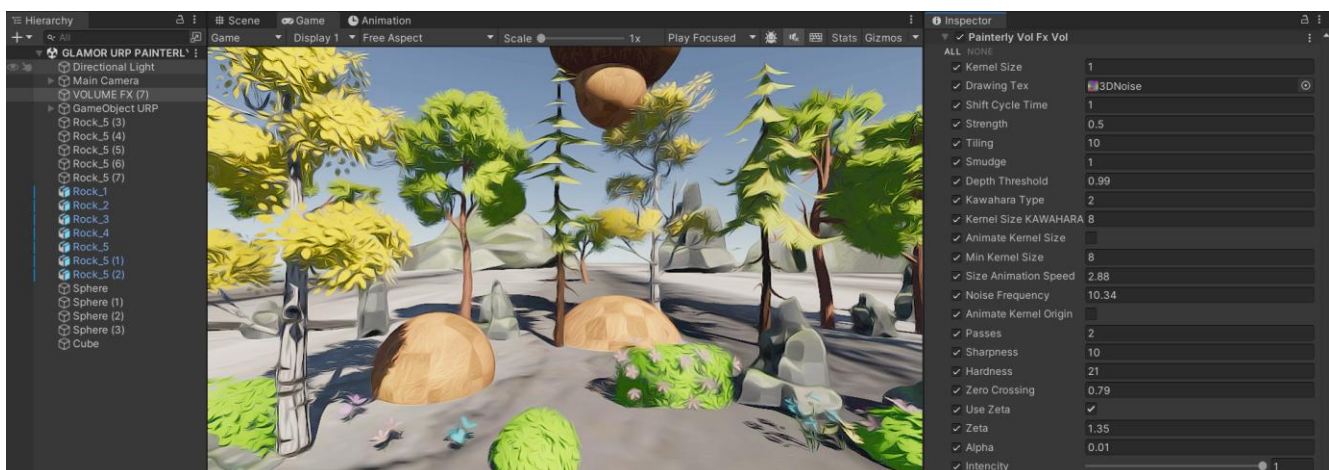
5. Bloomy full screen Haze



6. Screen space shadows

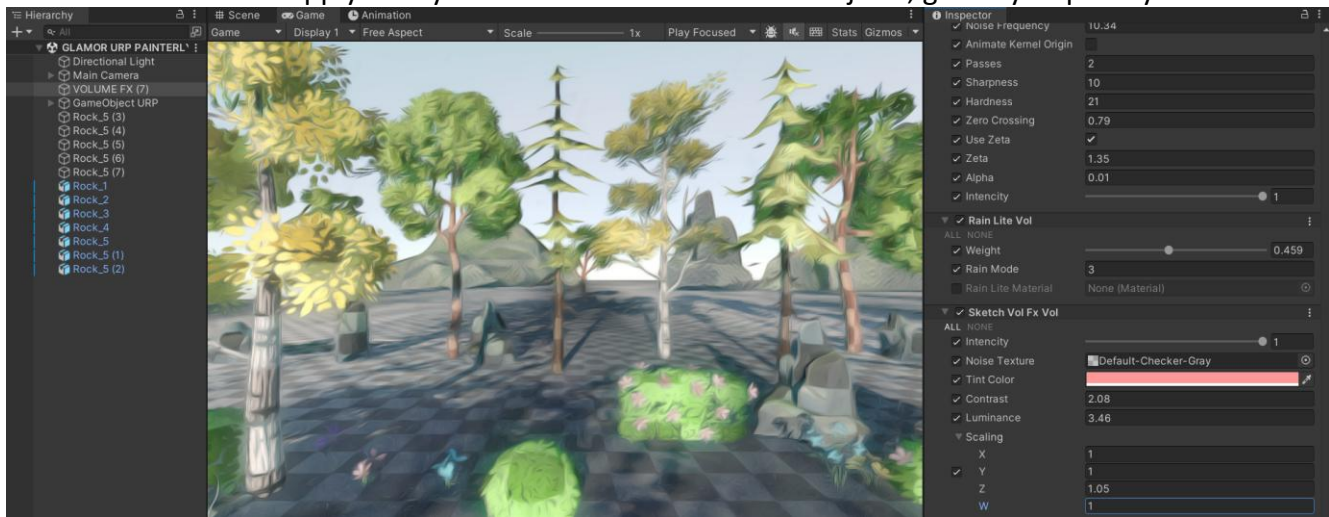


7. Kawahara painterly effect

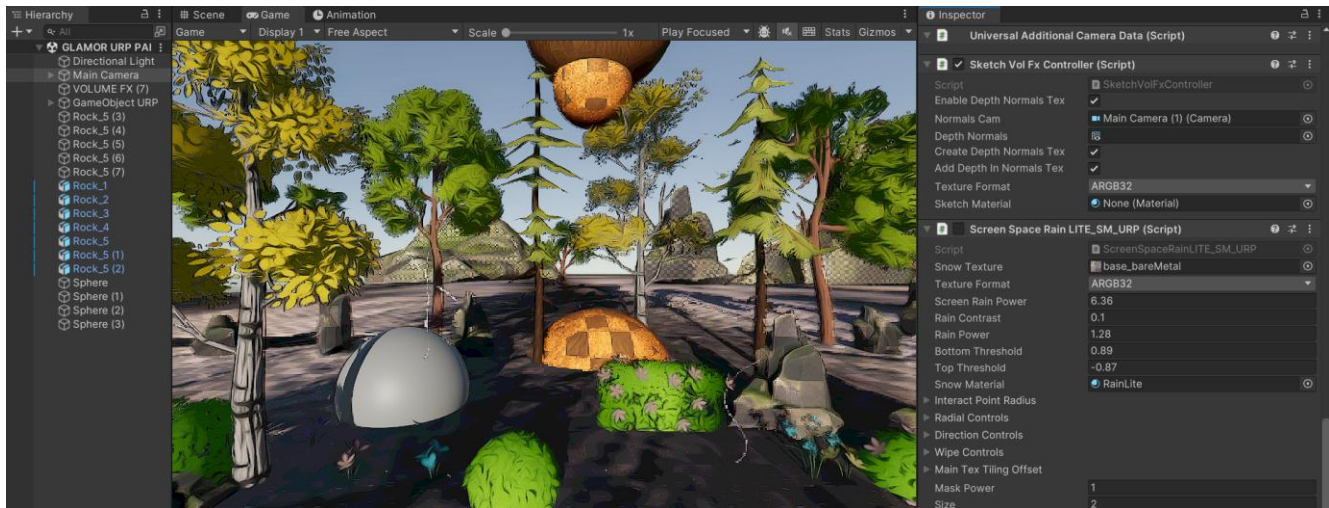


8. Sketch effect

The sketch effect can apply a fully stable texture on the scene objects, globally or per layer.

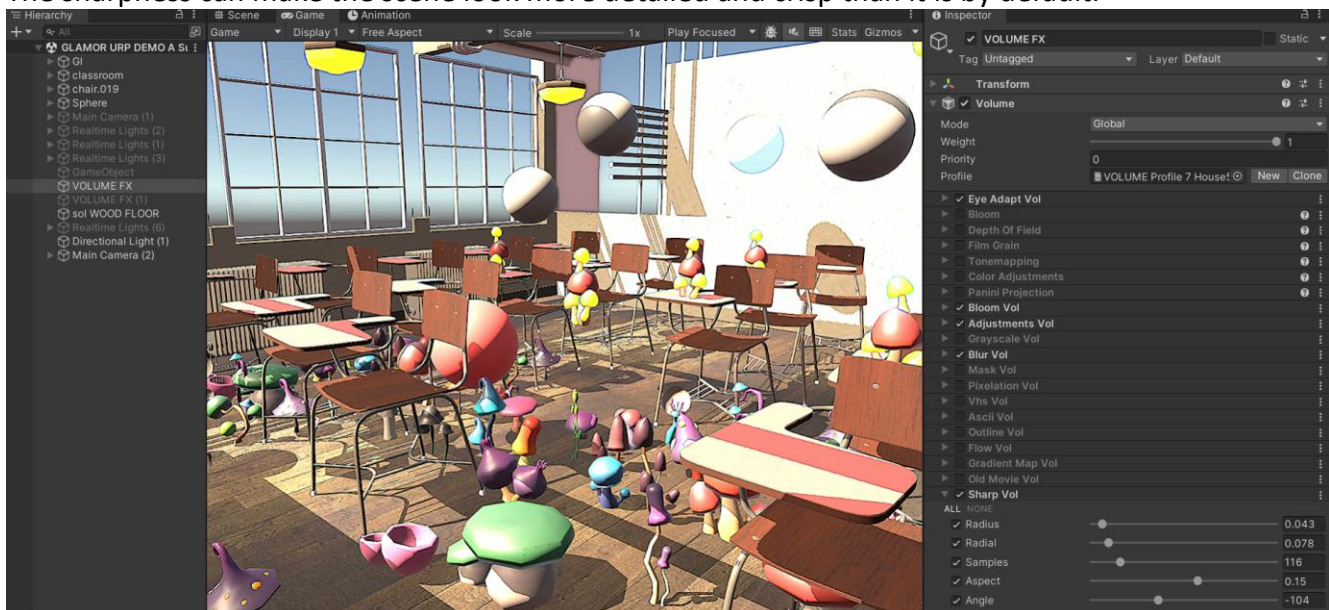


The sketch effect requires a Depth Normals texture, which can be enabled in the Rain Lite script controller or with the below shown “SketchVolFxController” script, attached on camera.

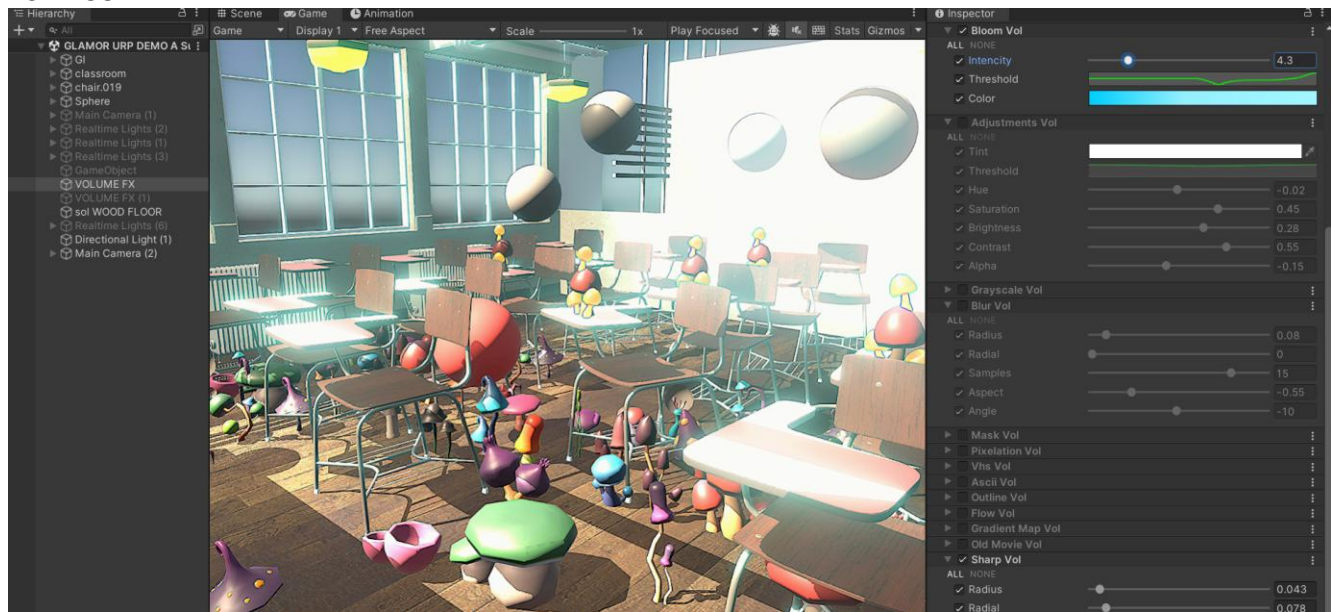


9. Sharpness effect

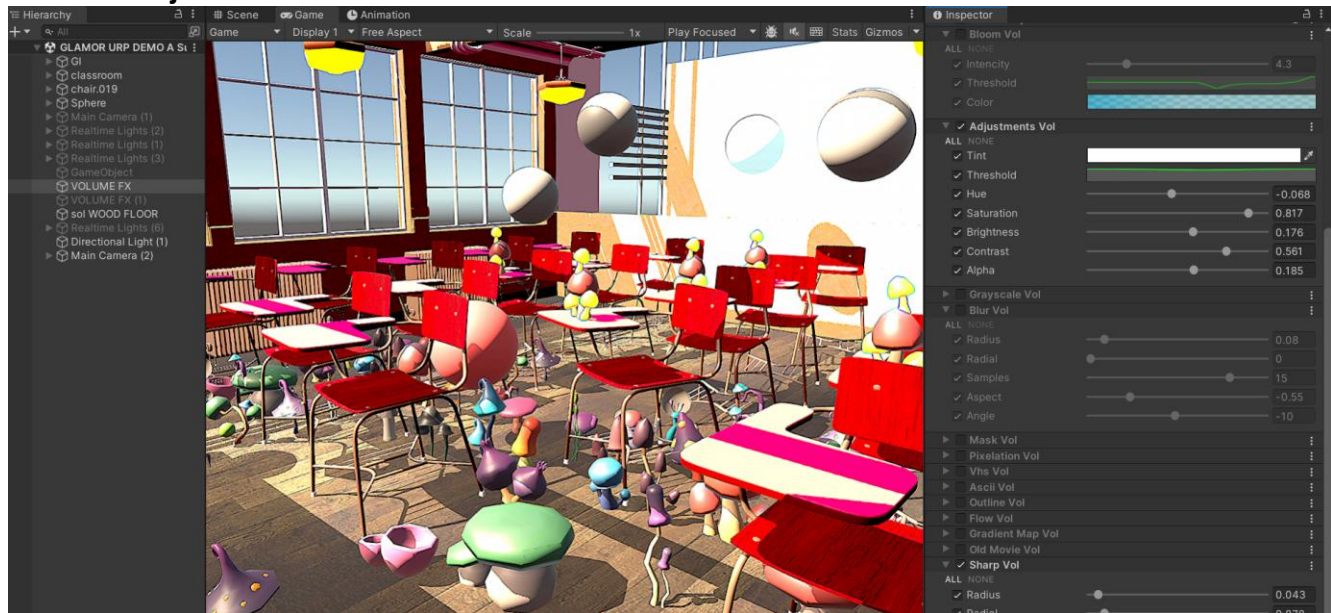
The sharpness can make the scene look more detailed and crisp than it is by default.



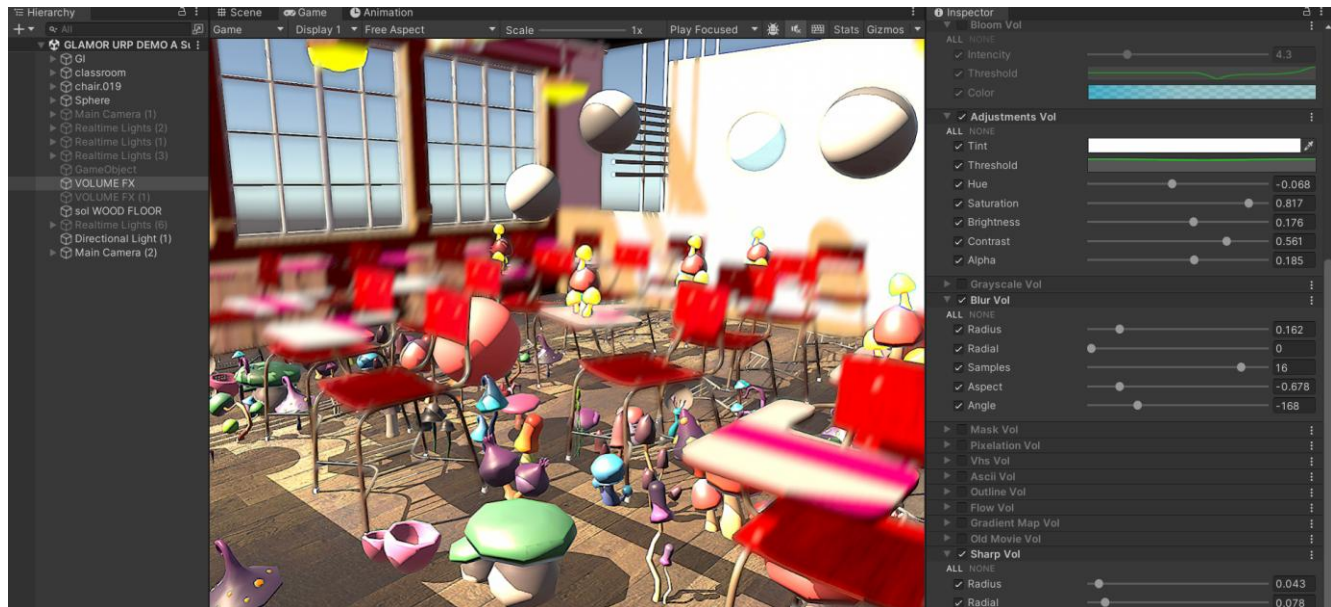
10. Bloom



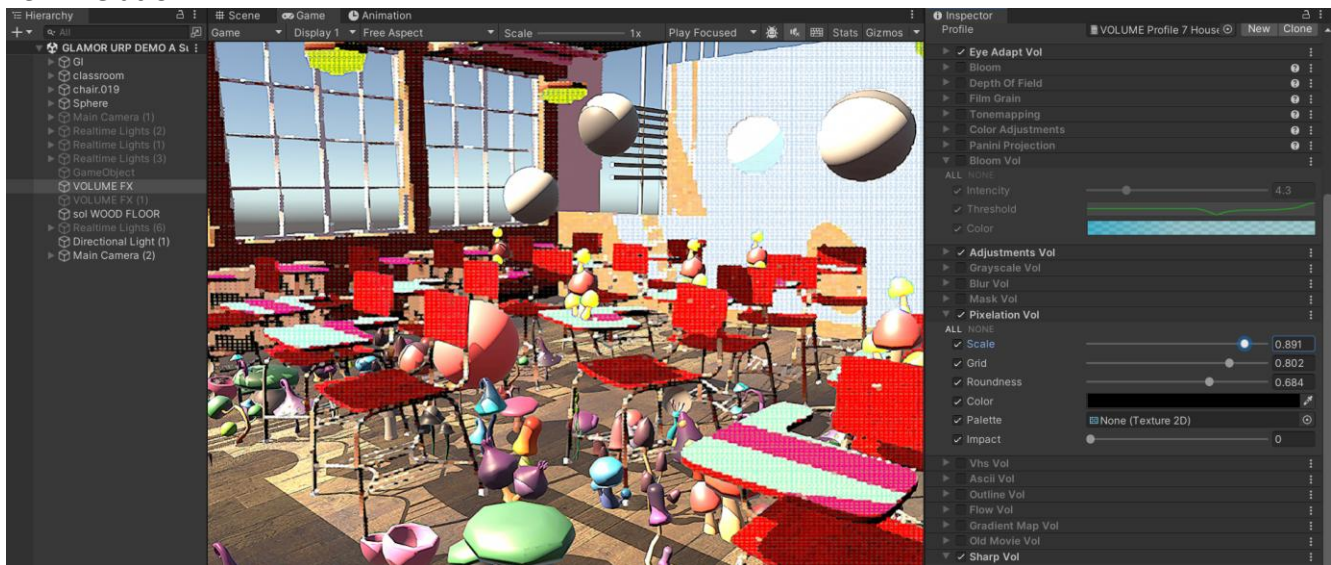
11. Color Adjustments



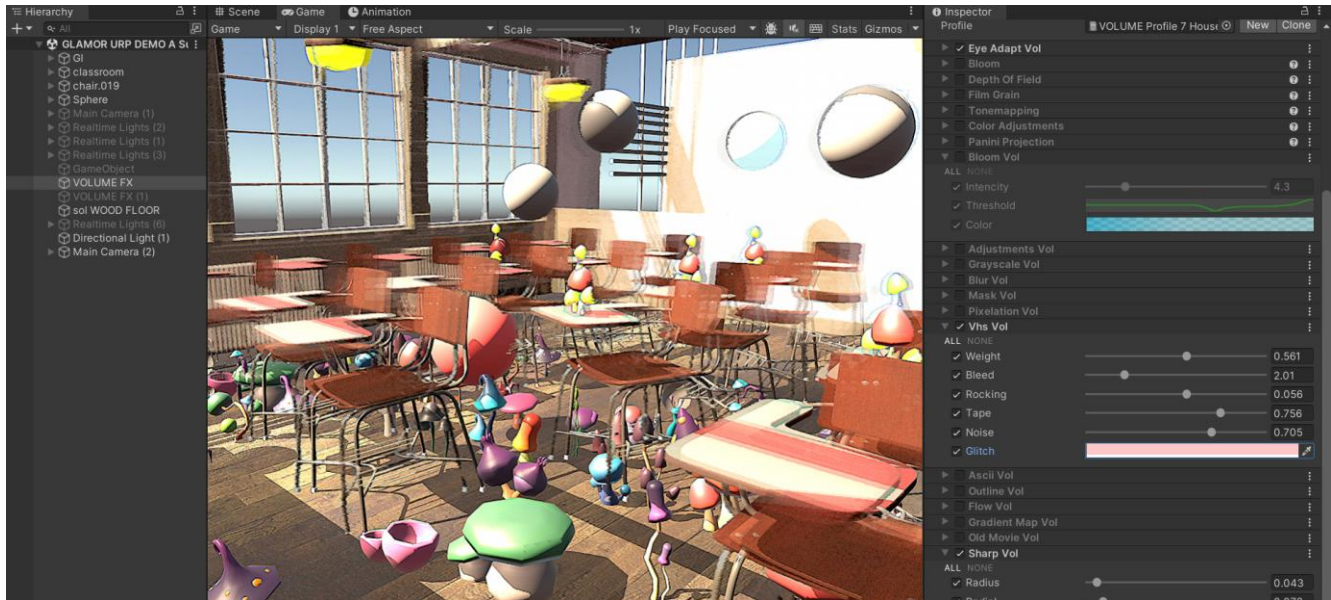
12. Blur



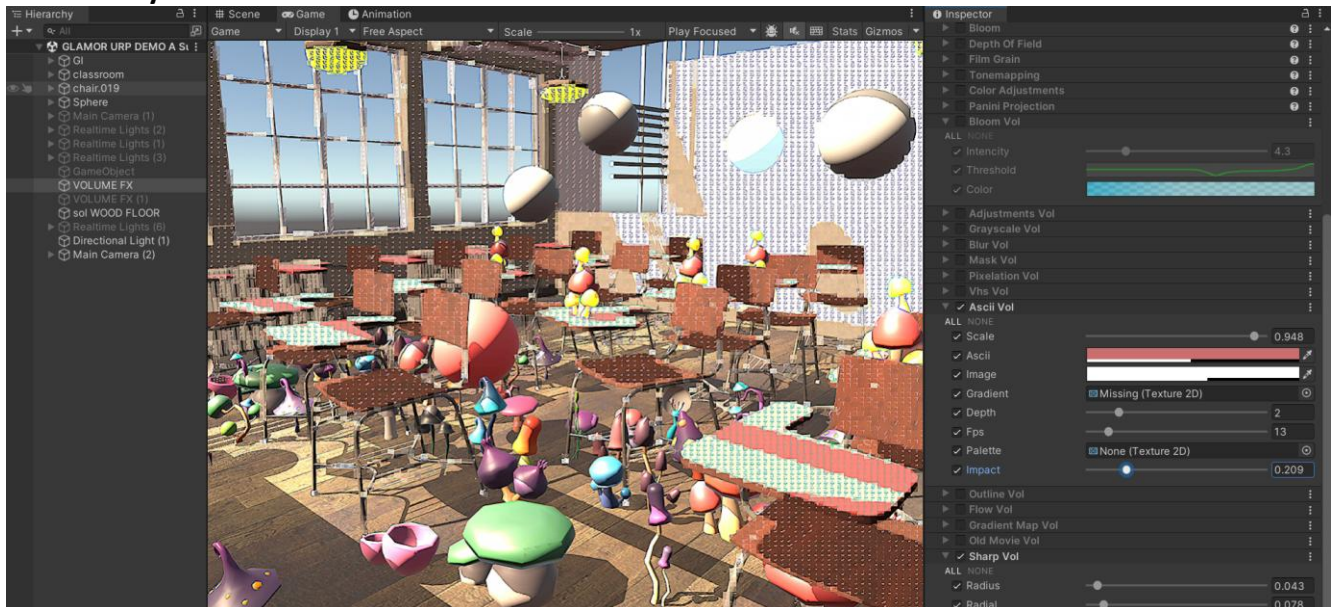
13. Pixelation



14. Vhs effect

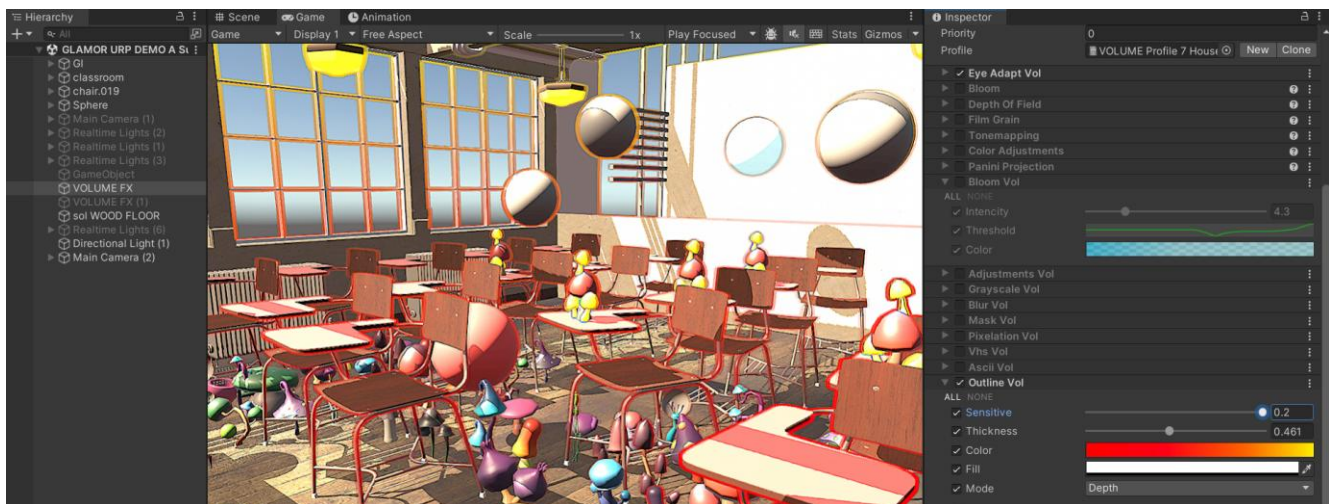
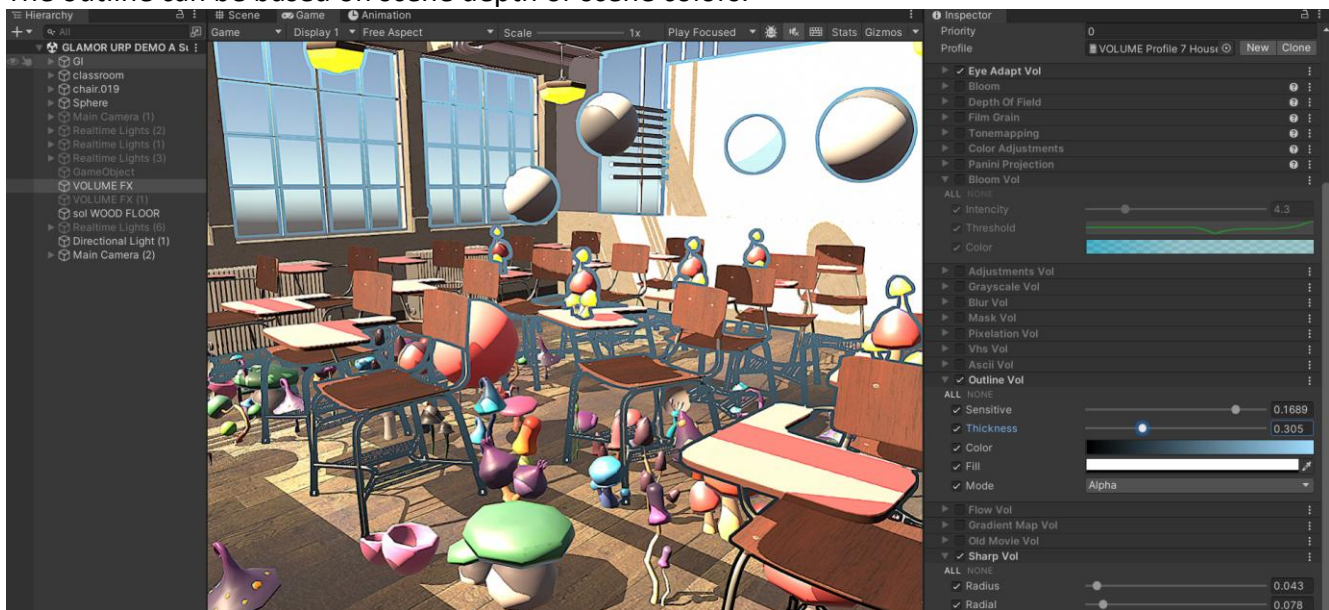


15. Ascii style

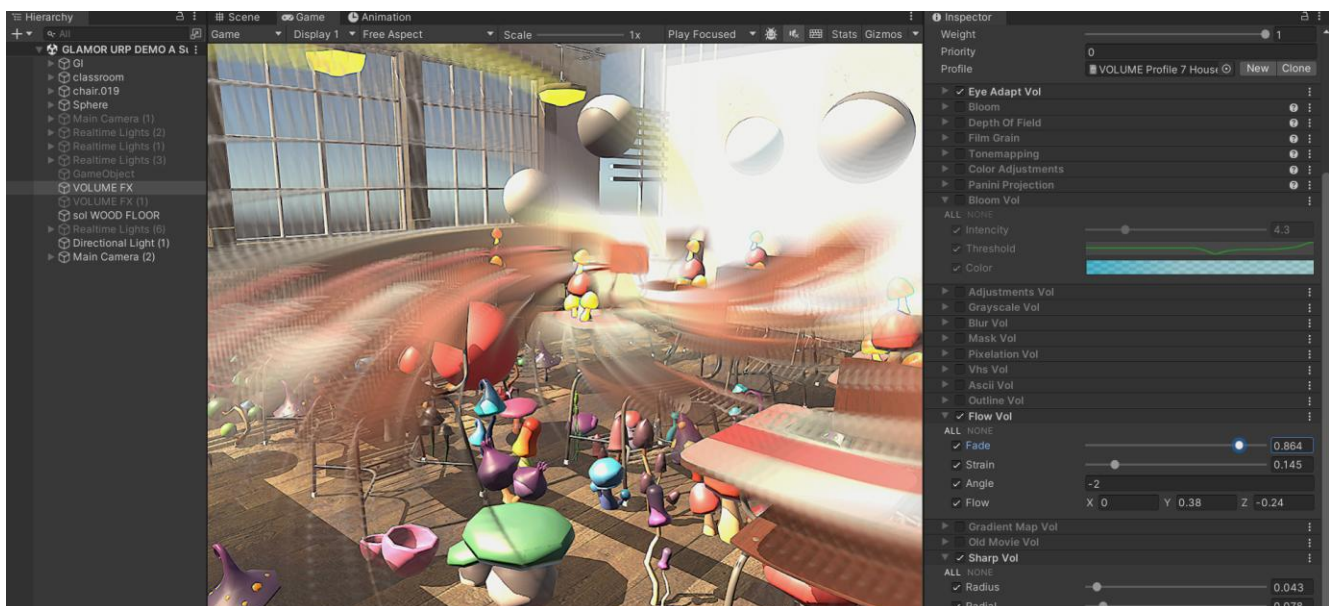


16. Outline effect

The outline can be based on scene depth or scene colors.



17. Flow effect



18. Gradient Mapping



19. Grayscale effect



20. Old movie effect

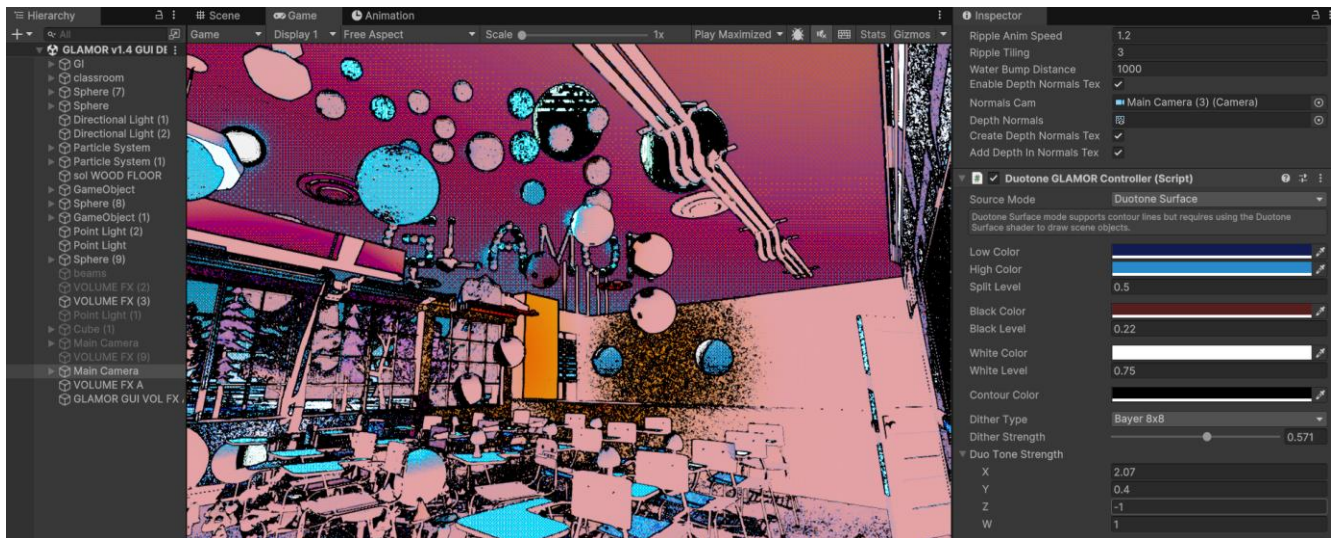


Standalone GLAMOR effects

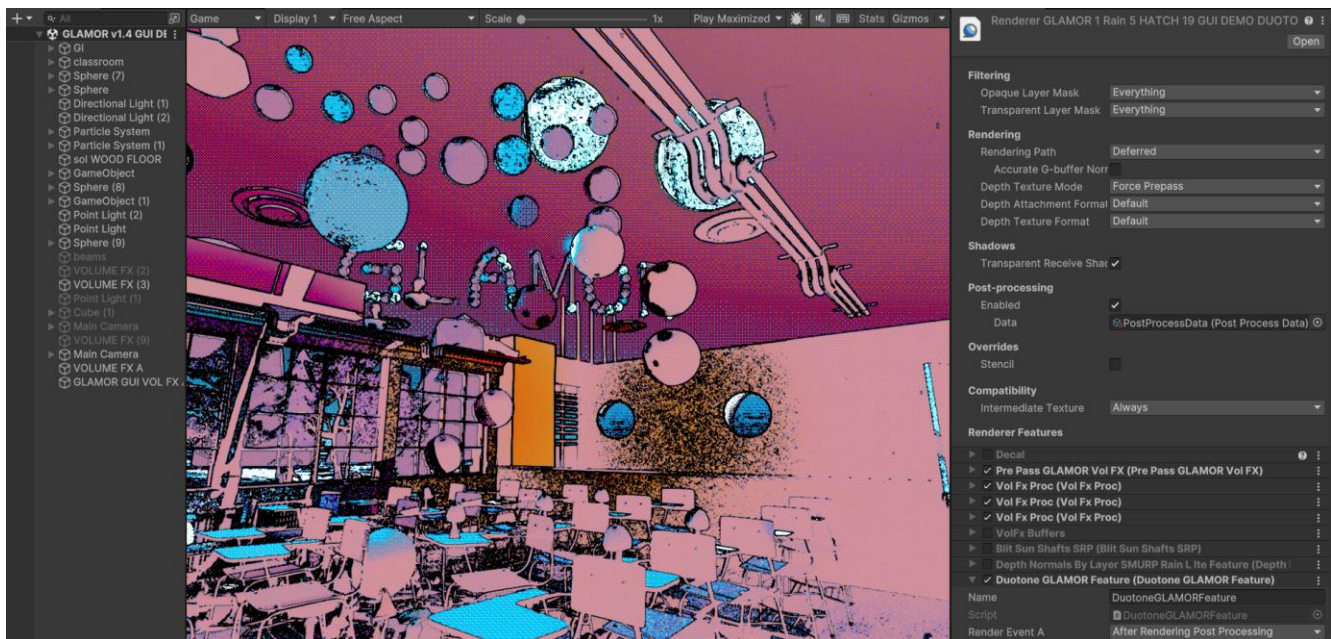
The system will be extended with standalone effects that can be used outside the VolFX framework, as a separate renderer feature in the Main Camera renderer.

1. Duo Tone effect

The first bonus effect in **GLAMOR v1.7** is a new duo tone effect for Unity 6 RenderGraph. The module uses the following control script on camera, **“DuoTone GLAMOR Controller”**.

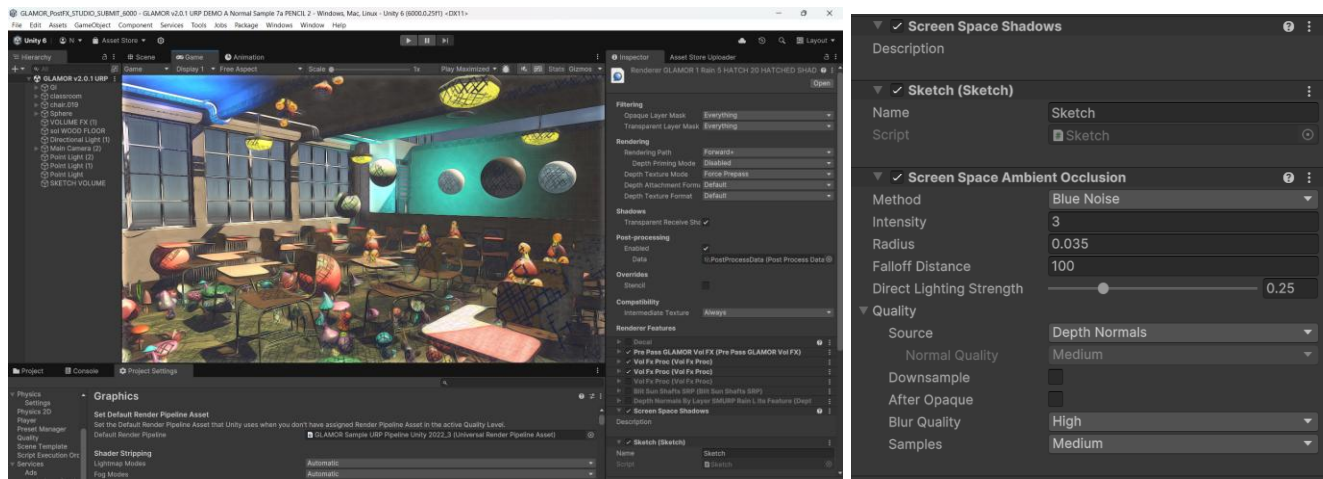


For the setup of the effect in the render, the **“DuoTone GLAMOR Feature”** must be inserted in the renderer of the camera as shown below.

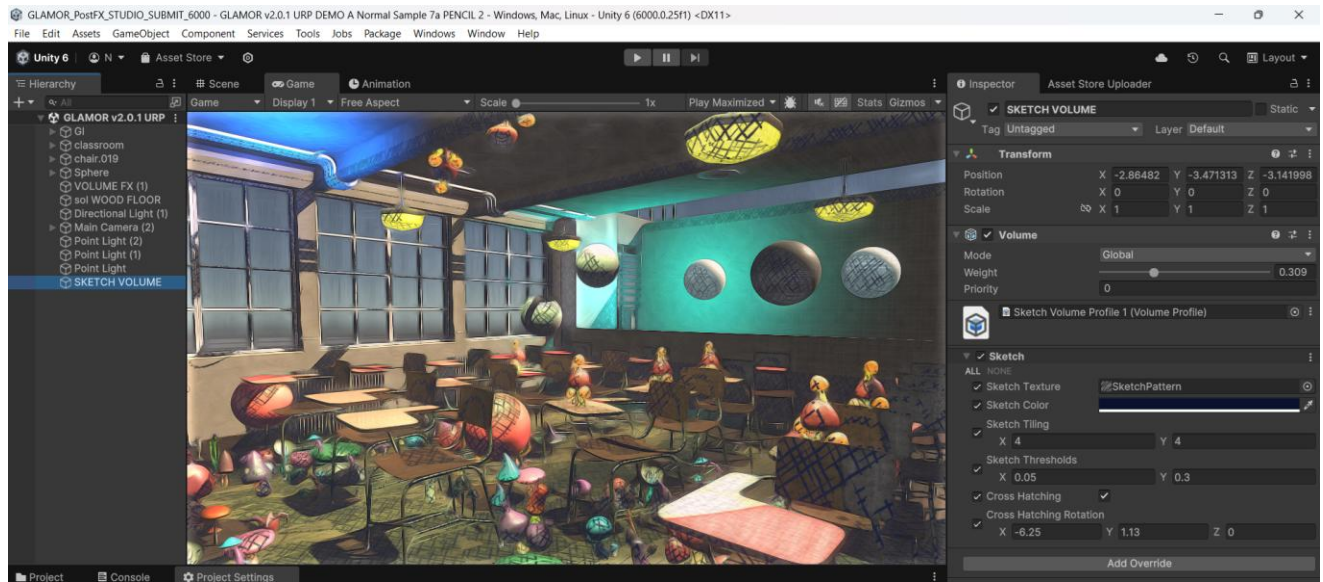


2. Hatched Shadows effect

The second bonus effect added in **GLAMOR v2.0** is a new screen space **Hatched Shadows** effect for Unity 6 RenderGraph. The effect **requires three renderer features**, Unity's own **Ambient Occlusion** and **Screen Space Shadows** and the **"Sketch"** renderer feature. An example renderer and demo scene are provided for direct reference and use.



To enable the effect in the scene an entry named **"Sketch"** can be added in the scene volume effects stack as shown below. The **Thresholds** values can vary the effect between shadow and light regions and fade it based on region. The cross hatch can add a second hatch pattern and its rotation can be controlled using the **Cross Hatch Rotation** values.



GLAMOR Known Issues

1. Use of GLAMOR with ODIN asset

Problem: Issue with entries missing or not be able to add effect entries in VolFX VFX renderer feature in GLAMOR when used with ODIN asset.

Solution: Open Odin Settings, go to editor types, uncheck the Artngame folder and click "update editors", then it won't update any of the inspector elements in that folder with the Odin counterparts.

[Video](#) of the setup for reference.

2. The system spams an error about not defined input axis in the console

This is an issue with the included sample camera controller that is using the legacy input system, this script is only for demo purposes and can either make use of the legacy input system in Unity settings to use it, or replace it with the Unity free camera script that is compatible with all Unity versions.

3. A warning is mentioned in the Unity log console about missing renderers at certain positions

This issue can appear in the included demo scenes if the sample GLAMOR pipeline that is used in the demo scenes is not properly set in the system. To resolve make sure to put the GLAMOR named pipeline in both Graphics and Quality Unity settings. The pipeline in Quality settings overrides the one defined in the Graphics.

GLAMOR Effects Combination

All effects can be combined in various ways, the render order is defined in the renderer VolFx entries by the ordering of the effects in the stack of the “VolFxProc” entries.

